



Building a Sample for Time Series Analysis

For the Inference of Macro Models

Course materials (slides, code, datasets): github.com/Vermandel/Macro-Modelling-Course-Nancy

This problem set is designed for a **one-hour in-class lab**. The goal is to build a small macroeconomic dataset for one country, transform the raw data into usable macro variables, and extract a business-cycle measure in MATLAB.

Instructions

Work in pairs if needed. Choose **one country** and collect the main quarterly macroeconomic series from the **OECD database via dbnomics**. Your final sample must contain at least:

- ▶ real GDP or GDP volume,
- ▶ one price index (CPI or GDP deflator),
- ▶ one short-term interest rate,

At the end of the session, you should be able to explain which series is in **price**, which series is in **volume**, and which transformation is needed before using the data in applied macroeconomics.

Exercise

1. Choose a country and download the data.

Use OECD quarterly data through `call_dbnomics.m`. Start from a GDP volume series, then find one price index, one interest rate, and one labor-market variable for the same country.

Some tips: downloading data

```
[output_mat,output_table,dates_nb] = ...  
    call_dbnomics('OECD/QNA/USA.B1_GE.VIXOBSA.Q');  
  
gdp_volume = output_mat(:,2);  
plot(dates_nb,gdp_volume)  
title('US real GDP')
```

Adapt the country code and the OECD series code to your own country.

2. Check carefully: price or quantity?

For each selected series, state whether it is:

- ▶ a **quantity/volume** variable,
- ▶ a **price** variable,
- ▶ a **rate** variable.

In particular:

- ▶ **GDP for real activity** should preferably be a **volume** measure, not current-price GDP;
- ▶ **inflation** should come from a **price index** such as CPI or the GDP deflator;
- ▶ **interest rates** and **unemployment rates** are already rates and should not be logged.

Briefly explain why using nominal GDP instead of real GDP would be problematic for business-cycle analysis.

3. Construct the main macroeconomic variables.

Create the following transformed series:

- ▶ output growth or log output,
- ▶ inflation,
- ▶ interest rate,
- ▶ unemployment rate.

You may use the following transformations:

$$g_t^y = 100 \times [\log(Y_t) - \log(Y_{t-1})],$$
$$\pi_t = 100 \times [\log(P_t) - \log(P_{t-1})].$$

Some tips: transformations

```
gdp_growth = 100*diff(log(gdp_volume));  
inflation   = 100*diff(log(cpi_index));
```

Comment on the difference between:

- ▶ the **level** of a price index,
- ▶ the **inflation rate** computed from this price index.

4. Check stationarity and extract the cycle.

Use at least **one** business-cycle filter on (log) real GDP, and if possible compare **two** filters.

Recommended MATLAB functions:

- ▶ **hpfiler**: Hodrick-Prescott filter.
- ▶ **bkfilter**: Baxter-King band-pass filter.

Some tips: HP and BK filters

HP filter (Smoothing=1600 for quarterly data):

```
log_gdp = log(gdp_volume);  
[trend_hp,cycle_hp] = hpfilter(log_gdp,Smoothing=1600);  
plot(dates_nb,cycle_hp); title('HP cycle of log GDP')
```

BK filter (cycles between 6 and 32 quarters):

```
[trend_bk,cycle_bk] = bkfilter(log_gdp, ...  
    LowerCutoff=6, UpperCutoff=32, LagLength=12);  
plot(dates_nb,cycle_bk); title('BK cycle of log GDP')
```

Note: `bkfilter` removes the first and last `LagLength` observations.

5. Interpret your results.

Answer the following short questions:

- (a) Which transformed variable would you use as the model counterpart of output?
- (b) Which transformed variable would you use for inflation?
- (c) Which filter gives the most convincing business-cycle measure for your country?
- (d) What practical data issue did you face: missing values, seasonal adjustment, code selection, or price/volume confusion?